Rutgers University
Math 151

Math 151 Previous Exams

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2025SP - Midterm Exam 1

1. Each part of this question is independent of the others. [Note: You do not need to show work on any part of this problem.]

   (a) Let $g(x) = \cos(x)$. If $(h \circ g)(x) = \cos^2(x) + 3\cos(x) - 8$, then find $h(x)$.

   (b) Find $\sin(\theta)$ where $\cos(\theta) = \frac{2}{5}$ and $\frac{3\pi}{2} < \theta < 2\pi$.

   (c) Find $\cos^{-1}\left(\cos\left(\frac{11\pi}{6}\right)\right)$.

   (d) Find $\lim_{x\to 1} f(x)$ where f is a function such that $f(1) = -5$ and

   $$\frac{x+3}{2} \leq f(x) \leq \sqrt{x^2+3} \quad \text{for} \quad x \neq 1.$$

   If there is Not Enough Information to find the limit, write NEI.

   (e) Consider the equation $x^3 + 2x + 1 = 0$. On which of the following intervals (if any) does the Intermediate Value Theorem guarantee that the equation has a solution? Fill in the bubble or bubbles containing correct answers.

   ○ $[-1, 0]$ ○ $[0, 1]$ ○ $[1, 2]$ ○ None of the Above

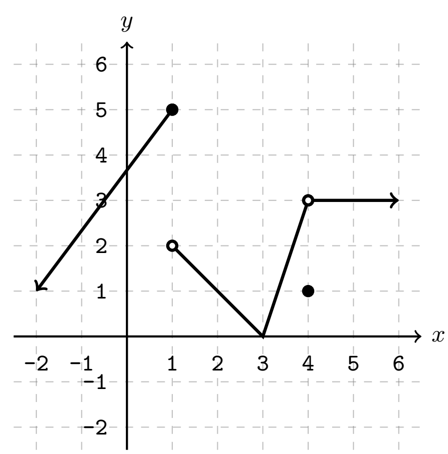
   (f) If $\lim_{x\to 2} \frac{p(x)+4}{x+1} = 8$, find $\lim_{x\to 2} p(x)$.

   (g) Find the equation(s) of the vertical asymptote(s) of $y = \frac{7x+7}{(x+1)(x+3)}$.

   (h) Simplify the following expression. Your answer may include w and t, but must not contain e or ln.

   $$\ln\left(\frac{e^{5w}}{e^{4t}}\right)$$

2. Consider the graph of the following function $f(x)$. Find each of the following. If one or more of them does not exist, write DNE.



   (a) $f(1)$

   (b) $\lim_{x\to 1^+} f(x)$

   (c) $\lim_{x\to 1} f(x)$

   (d) $f(4)$

   (e) $\lim_{x\to 4^+} f(x)$

   (f) $\lim_{x\to 4} f(x)$

3. Compute the following limits. If the limit is infinite, write $+\infty$ or $-\infty$ as appropriate. If the limit does not exist and is not infinite, write DNE as your answer. Show all work and use proper limit notation.

   (a) $\displaystyle\lim_{x\to 0} \frac{\sin(9x)}{\sin(2x)}$

   (b) $\displaystyle\lim_{x\to 2} \frac{\frac{1}{3} - \frac{1}{x+1}}{x-2}$

   (c) $\displaystyle\lim_{x\to -\infty} \frac{\sqrt{49x^2+4}}{x+3}$

   (d) $\displaystyle\lim_{x\to 5^+} \frac{25-x^2}{x^2-10x+25}$

4. Find the values A and B such that $f(x)$ is continuous for all real numbers x, where

$$f(x) = \begin{cases} \dfrac{7x-7}{|x-1|}, & x < 1 \\ A, & 1 \le x \le 3 \\ \ln(x+B), & x > 3 \end{cases}$$

   Show all work and use limits to justify your answers. Answers that are not justified by limits will not receive full credit.

5. Find equations for all horizontal and vertical asymptote(s) of the function

$$f(x) = \frac{3e^x + 8}{7 - e^x}.$$

   Each asymptote that you find must be supported by work based on limits.

6. Let $f(x) = \sqrt{x-2}$. Use this information to answer parts (a)–(c).

   (a) Use the definition of the derivative to write $f'(11)$ as a limit. Do not evaluate the limit until part (b).

   (b) *Now* evaluate the limit you wrote in part (a) to find $f'(11)$.

   (c) Use your answer to part (b) to find an equation of the line tangent to $y = f(x)$ at $x = 11$.

2025SP - Midterm Exam 2

1. For each part, calculate the derivative. Do not simplify your answers. Each part of this question is graded **right/wrong.**

   (a) $\dfrac{d}{dx}\left(\dfrac{4}{x^3} - 6\sqrt[5]{x} + 11\cos(5)\right)$

   (b) $\dfrac{d}{dx}\left(\dfrac{2\ln(x)}{3^x - 1}\right)$

   (c) $\dfrac{d}{dx}\left(\tan\left(e^{7x^2-2}\right)\right)$

   (d) $\dfrac{d}{dx}\left(x\sin^{-1}(4x)\right)$

2. Suppose $f(2) = 11$, $f'(2) = -12$, and $g(x) = x - f(\sqrt{x})$. You may assume that f is invertible. Each part of this question is graded **right/wrong**.

   (a) Find an equation of the line tangent to $y = f(x)$ at $x = 2$.

   (b) Find an equation of the line tangent to $y = f^{-1}(x)$ at $x = 11$.

   (c) Find an equation of the line tangent to $y = g(x)$ at $x = 4$.

3. Let $\mathcal{C}$ be the curve defined by the equation $x^2 - 5xy + 25y^2 = 12$.

   (a) Find $\dfrac{dy}{dx}$.

   (b) Find the point(s) (x, y) on the curve $\mathcal{C}$ where the line tangent to the curve is horizontal.

4. A model rocket is shot straight up into the air and falls straight back down to the ground. Its height (in feet) above the ground from the time it is launched to the time it hits the ground is given by the following function:

$$s(t) = 70t - 5t^2$$

   after t seconds.

   (a) Find the function $v(t)$ that gives the *velocity* of the rocket and the function $a(t)$ that gives the *acceleration* of the rocket after t seconds.

   (b) When does the rocket reach its maximum height?

   (c) On what time interval is the rocket traveling upward? Give your answer in interval notation.

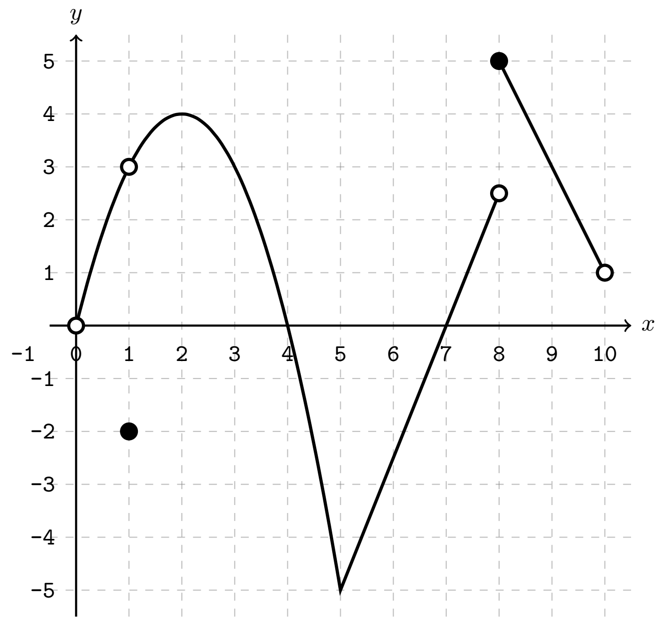
5. Let$f(x) = (5x + 8)^x$. (Note that x is an exponent.) Find $f'(x)$. Your final answer should be in terms of x only.

6. Find the constants A and B so that the function $f(x) = (Ax + B)e^{x-2}$ has a tangent line with a slope of 7 at the point $(2, 4)$.

7. The function $f(x)$ shown in the graph on the right has domain $(0, 10)$. Use this graph to answer the following questions.

   If there is not enough information to answer a question, write NEI.

   Partial credit is possible on this problem, but we will only grade what is in the answer boxes.



   (a) For which values of x in the interval $(0, 10)$ is $f(x)$ not continuous?
   (b) For which values of x in the interval $(0, 10)$ is $f(x)$ not differentiable?
   (c) Find the values of x in the interval $(0, 10)$ for which $f'(x) = 0$.
   (d) Where is $f'(x)$ negative? Use interval notation for your answer.
   (e) Find the value of $f'(9)$.

8. A spherical balloon filled with helium is deflating at a rate of 144 cm^3 per hour. Let V be the volume of the sphere and let r be its radius. [Hint: The volume of a sphere is given by the formula: $V = \frac{4}{3}\pi r^3$.]

   (a) What is the value of $\frac{dV}{dt}$? Your answer should be a number, not a formula.
   (b) Find the rate at which the radius of the balloon is changing at the moment when the radius is 3 cm.

2025SP - Midterm Exam 3

1. Let $f(x) = \sqrt{x}$. Use this function in parts (a)-(b) below.
   (a) Find the linearization of $f(x)$ at $a = 36$.
   (b) Use the linearization in part a) to *estimate* $f(34)$. Write your final answer as a simplified fraction.

2. Evaluate the following indefinite integrals.
   (a) $\displaystyle\int \left(\cos(3x) - \frac{7}{x^2+1}\right) dx$
   (b) $\displaystyle\int \frac{7+13x^6}{x}\, dx$

3. The parts of this question are independent.
   (a) Let $g(x) = (x-2)^{1/3}$. On which on the following intervals does $g(x)$ satisfy the hypotheses of the Mean Value Theorem? (There is only one correct answer.)
   A. $[0, 3]$ B. $[-1, 2]$ C. $[-2, 5]$ D. $[1, 4]$ E. None of the above
   (b) Suppose h is a differentiable function with $h'(x) \le 5$, and $h(4) = -9$. What is the maximum possible value of $h(7)$?

4. An object's acceleration is given by $a(t) = e^{-t} + 6$. Find the velocity $v(t)$ and the position $s(t)$ of the object if $v(0) = 3$ and $s(0) = 5$.

5. Let $f(x) = -4x + 8\sin(x)$. Find the absolute extrema and where they occur for $f(x)$ on the interval $[0, \pi]$. Your answers must be *exact*, but you might find the approximations $\pi \approx 3.1$ and $\sqrt{3} \approx 1.7$ useful to determine the correct answers.

6. A function $f(x)$ and its first and second derivatives are given as follows:

$$f(x) = \frac{x-5}{(x-3)^3} \qquad f'(x) = -\frac{2(x-6)}{(x-3)^4} \qquad f''(x) = \frac{6(x-7)}{(x-3)^5}$$

Find all of the following or write NONE as appropriate. If an answer has multiple open intervals, separate them with commas. If there is not enough information to find an answer, write NEI in the box. We will only grade what is written in the answer box on this problem.

Open interval(s) where f is increasing:

Open interval(s) where f is decreasing:

x-coordinate(s) of local maximum value(s):

x-coordinate(s) of local minimum value(s):

Open interval(s) where f is concave up:

Open interval(s) where f is concave down:

x-coordinate(s) of inflection point(s):

Equation(s) of vertical asymptote(s):

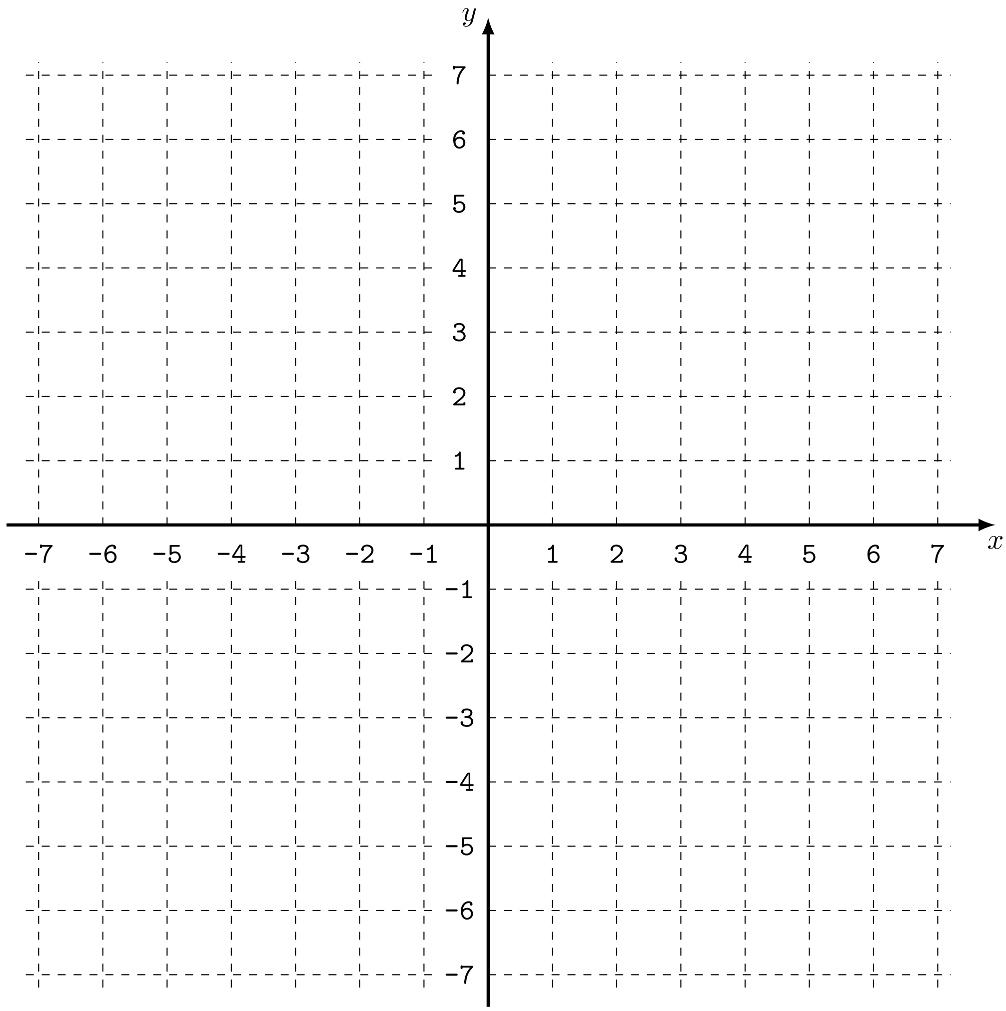
Equation(s) of horizontal asymptote(s):

7. Sketch the graph of a function $f(x)$ with the following properties:

- f is continuous on $(-\infty, \infty)$ and $y = 2$ is an asymptote of f
- $\lim_{x \to -\infty} f(x) = \infty$
- $f(-3) = -2$, $f(-1) = 1$, $f(2) = 5$ and $f(4) = 3$
- The first and second derivatives of f have the following sign chart:

x	$(-\infty, -3)$	$(-3, -1)$	$(-1, 2)$	$(2, 4)$	$(4, \infty)$
$f'(x)$	$-$	$+$	$+$	$-$	$-$
$f''(x)$	$+$	$+$	$-$	$-$	$+$

Label the asymptote, and all local extrema and inflection points.



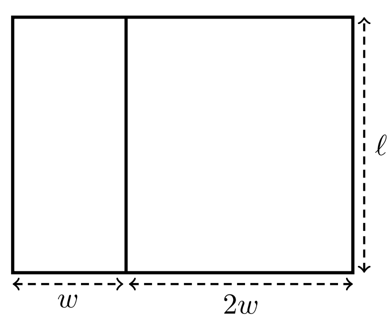
8. Calculate the following limits. Show all work and use proper limit notation. If you use L'Hôpital's Rule, indicate where and say which indeterminate form allows its use.

   (a) $\lim_{x \to 0} \dfrac{e^{9x} - 1 - 9x}{x^3 + 4x^2}$

   (b) $\lim_{x \to 0} (1 + \sin(5x))^{1/x}$

9. Farmer Brown wants to enclose two adjacent pens with the minimal amount of fencing possible. There are two constraints on the pens:

   - The two pens have the same length ℓ, but the second one is twice as wide as the first one. (The width of the first pen is denoted by w, see the figure on the right.)
   - The total area enclosed by the two pens is 150 m^2.



(a) Write a formula for $F(w)$, the amount of fencing used in terms of w only.

(b) Find the values of w and ℓ that produce the minimum amount of fencing. You must use calculus to justify that you have found a minimum.

2024SP - Midterm Exam 1

1. Right/wrong questions.

   (a) Let $g(x) = \sqrt{x} + 1$. If $(h \circ g)(x) = \dfrac{7}{\sqrt{x} + 6} + \sin(\sqrt{x} + 1)$, then find $h(x)$.

   (b) Find $\cos\theta$ where $\sin\theta = -2/9$ and $\frac{3\pi}{2} < \theta < 2\pi$.

   (c) Find $\sin^{-1}\left(\sin\left(\frac{5\pi}{4}\right)\right)$.

   (d) If $\sqrt{10 - x^2} \le f(x) \le (2 + x)$ for $-2 \le x \le 2$, find $\lim_{x\to 1} f(x)$. If there is Not Enough Information to find the limit, write NEI.

   (e) Find $\sin(t)$ where $\tan(t) = -5/3$ and $\cos(t) < 0$.

   (f) If $\lim_{x\to 1} \dfrac{p(x) - 3}{x + 4} = 2$, find $\lim_{x\to 1} p(x)$.

   (g) Find the equation(s) of the vertical asymptote(s) of $y = \dfrac{x + 1}{(x + 4)(7x + 7)}$.

   (h) Find the following limit: $\lim_{x\to 3^+} \dfrac{x^2 - 9x + 14}{x^2 + 3x - 18}$. If it is infinite write ∞ or $-\infty$ as appropriate, write DNE if it does not exist *and* is not infinite and write NEI if there is not enough information to find it.

2. Evaluate $\lim_{x\to 9} \dfrac{x^2 - 3x - 54}{\sqrt{3x - 2} - 5}$.

3. Find all horizontal and vertical asymptote(s) of the function $f(x) = \dfrac{\sqrt{4x^2 + 5}}{2x + 10}$.

4. Find the values A and B such that $f(x)$ is continuous for all real numbers x, where

$$f(x) = \begin{cases} \dfrac{A|x-2|}{x-2}, & x < 2 \\ 7, & 2 \le x \le 3 \\ \ln(Bx + 4), & x > 3 \end{cases}$$

   Use limits to justify your answers. Answers that are not justified by limits will not receive full credit.

5. A student calculated $\lim_{x\to 0} \dfrac{\sin(3x)}{\tan(4x)}$, but with some steps missing between line (X) and line (Y).

$$\lim_{x\to 0} \frac{\sin(3x)}{\tan(4x)} = \qquad \textbf{(X)}$$

$$(\cdots)$$

$$= \left(\lim_{x\to 0} \frac{\sin(3x)}{3x}\right) \cdot \left(\lim_{x\to 0} \frac{4x}{\sin(4x)}\right) \cdot \left(\lim_{x\to 0} \frac{3x}{4x}\right) \cdot \left(\lim_{x\to 0} \cos(4x)\right) \qquad \textbf{(Y)}$$

(a) Add one or more missing steps to show how to get from line (X) to line (Y):

(b) Find $\lim\limits_{x \to 0} \dfrac{\sin(3x)}{\tan(4x)}$ by evaluating the expression in line (Y).

6. Let $f(x) = \dfrac{1}{x-7}$. Use this information to answer parts (a) and (b).

   (a) Use the limit definition of the derivative to evaluate $f'(1)$, and then write its value in the box below.

   (b) Use your answer to part (a) to find an equation of the line tangent to $y = f(x)$ at $x = 1$.

7. Use the Intermediate Value Theorem (IVT) to show that the equation

$$x^3 - 18x + 1 = 0$$

   has a solution in the interval $[0, 1]$. Do not find the solution. Explain how you used the theorem in a single, concise, English sentence.

8. Sketch the graph of a function that meets the following conditions:

   - $\lim\limits_{x \to -\infty} f(x) = 4$
   - $\lim\limits_{x \to -3^-} f(x) = +\infty$ and $\lim\limits_{x \to -3^+} f(x) = +\infty$
   - $f(0) = 1$
   - $\lim\limits_{x \to 2^-} f(x) = +\infty$ and $\lim\limits_{x \to 2^+} f(x) = -\infty$
   - $y = -5$ is a horizontal asymptote of $f(x)$

   In your sketch, indicate all asymptotes with dotted lines labeled by their equations.

2024SP - Midterm Exam 2

1. Suppose $f(3) = 7$, $f'(3) = 2$, and $g(x) = x^4 f(3x)$. You may assume that f is invertible.
   (a) Find an equation of the line tangent to $y = f(x)$ at $x = 3$.
   (b) Find an equation of the line tangent to $y = f^{-1}(x)$ at $x = 7$.
   (c) Find an equation of the line tangent to $y = g(x)$ at $x = 1$.

2. For each part, calculate the derivative. Do not simplify your answers.
   (a) $\dfrac{d}{dx}\left(\dfrac{4}{x^2} - 7\sqrt[4]{x} + 10\ln x\right)$
   (b) $\dfrac{d}{dx}\left(2x^5 \tan x\right)$
   (c) $\dfrac{d}{dx}\left(\sin\left(\dfrac{x+3}{5^x}\right)\right)$
   (d) $\dfrac{d}{dx}\left(4\sin^{-1}\left(e^x\right)\right)$

3. Consider the following function:

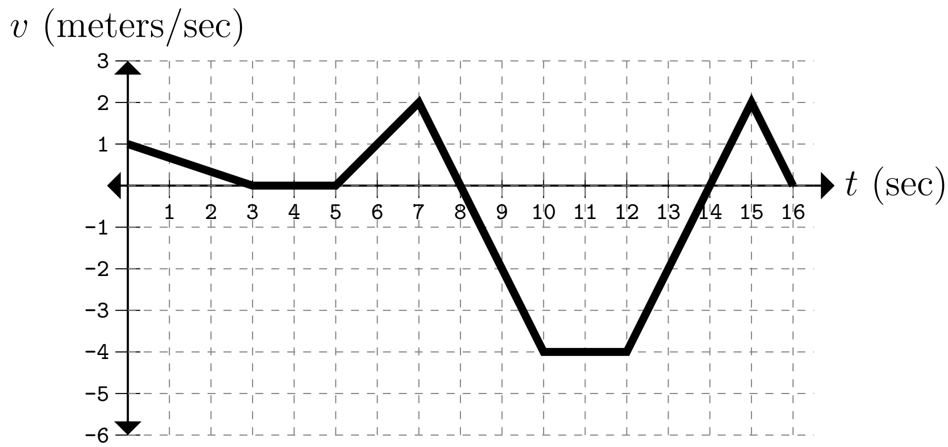
$$f(x) = \begin{cases} x^2 - 6x + 2 & x \leq 2 \\ Ax + B & x > 2 \end{cases}$$

   Find the values of A and B that make the function differentiable at $x = 2$.

4. Let $\mathcal{C}$ be the curve defined by the equation $4x^2 + 9y^2 + 6xy = 108$.
   (a) Find $\dfrac{dy}{dx}$.
   (b) Find the point(s) (x, y) on the curve $\mathcal{C}$ where the line tangent to the curve is horizontal.
   (c) Find the point(s) (x, y) on the curve $\mathcal{C}$ where the line tangent to the curve is vertical.

5. The figure below shows the velocity $v = f(t)$ of a particle moving on a horizontal coordinate line for the time interval $0 \leq t \leq 16$ seconds. In the coordinate line in which the particle moves, the positive direction is *to the right.*

Velocity as a Function of Time



For each question below, give your answers as specific times (e.g., $t = 5$), time intervals (e.g., $2 < t < 8$), or combinations of these, as applicable.

(a) When is the particle is moving to the *left*?

(b) When is the particle moving at its greatest speed?

(c) When is the particle's acceleration negative?

(d) What is the particle's exact acceleration at $t = 14$?

(e) What is the particle's average acceleration on the interval $3 \le t \le 7$?

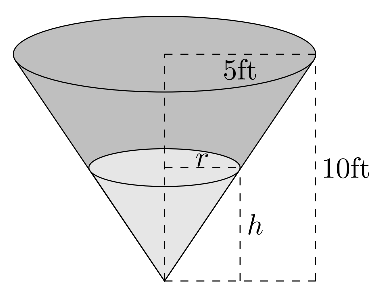
6. Let $f(x) = x^{7x^2+4}$ on the domain $x > 1$. Find $f'(x)$. (Note that $7x^2 + 4$ is an exponent.) Your final answer should be in terms of x.

7. Let $y = \sin^{-1}(x)$. Apply implicit differentiation to the equation:

$$\sin(y) = x$$

and use trigonometric methods (such as an identity or a right triangle) to find an expression for $\dfrac{dy}{dx}$ entirely in terms of x. Show all of your work. Write your final answer in the box provided.

8. Water runs into a large conical tank at the rate of 12ft^3 per minute. The tank has a height of 10 feet and a radius of 5 feet. The water is also shaped like a cone, and it has a radius of r feet and a height of h feet. Please see the figure to the right. [Hint: The volume of a cone of radius R and height H is given by $V = \frac{1}{3}\pi R^2 H$]



(a) What is the value of $\dfrac{dV}{dt}$? Your answer should be a number, not a formula.

(b) Use similar triangles to write h in terms of r.

(c) At what rate is the height of the water changing when the height is 8ft? Use correct units in your answer.

2024SP - Midterm Exam 3

1. The parts of this question are independent.

   (a) Find the indefinite integral $\int \left(x^6 + \cos(5x)\right) dx$

   (b) An object's acceleration is given by $a(t) = \sin(t) + 4$ for $t > 0$. Find the velocity $v(t)$ and the position $s(t)$ of the object for $t > 0$, if $v(0) = -2$ and $s(0) = 1$.

2. The parts of this question are independent.

   (a) Let $f(x)$ be a function continuous on the interval $[0, 3]$ and differentiable on the interval $(0, 3)$. This function also has the property that $f(0) = f(3)$. Use the Mean Value Theorem to determine which of the following sentences is true and fill in the circle next to the true statement.

   ◯ The graph of $f(x)$ *must* have an x-intercept on the interval $(0, 3)$.

   ◯ $f(x)$ *must* have a critical point on the interval $(0, 3)$.

   ◯ $f(x)$ *must* have both an x-intercept *and* a critical point on the interval $(0, 3)$.

   ◯ It is possible that $f(x)$ has neither an x-intercept nor a critical point on the interval $(0, 3)$.

   (b) Suppose g is a differentiable function with $g'(x) \leq 2$, and $g(2) = 5$. What is the maximum possible value of $g(5)$?

3. Let $f(x) = e^{-x^2}$. Find the absolute extrema and where they occur for $f(x)$ on the interval $[-2, 1]$.

4. Let $f(x) = \dfrac{x^2 + 6x + 9}{x + 5}$. You may use that $f'(x) = \dfrac{x^2 + 10x + 21}{(x + 5)^2}$. Find the open interval(s) where f is increasing and where f is decreasing, and the x-coordinate(s) of any local extrema.

5. Let f be a twice differentiable function, and suppose its second derivative is given by

$$f''(x) = 12\sin(x)\cos(x) \quad \text{for } x \text{ in } (-\pi, \pi).$$

   Part (a) and (b) below relate to the function f.

   (a) The point $x = -\pi/4$ is a critical point of f. Is $x = -\pi/4$ the location of a local maximum, local minimum, neither, or is it impossible to tell from the information given?

   (b) Find the open interval(s) on $(-\pi, \pi)$ where f is concave up and where f is concave down, and the x-coordinate(s) of inflection point(s).

6. Sketch the graph of a function $f(x)$ with the following properties:

   - The domain of $f(x)$ is $(-\infty, -2) \cup (-2, \infty)$ and $x = -2$ is an asymptote of f
   - $f(-4) = 3$, $f(0) = 4$, $f(1) = 1$, and $f(3) = -2$

- The first and second derivatives of f have the following sign chart:

x	$(-\infty, -4)$	$(-4, -2)$	$(-2, 0)$	$(0, 1)$	$(1, 3)$	$(3, \infty)$
$f'(x)$	$-$	$+$	$+$	$-$	$-$	$+$
$f''(x)$	$+$	$+$	$-$	$-$	$+$	$+$

Label the asymptote, and all local extrema and inflection points.

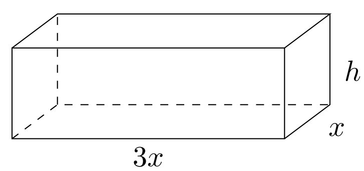
7. Calculate the following limits:

   (a) $\lim_{x \to 0} \left(\frac{1 - \cos(x)}{6x^2 - x^3} \right)$

   (b) $\lim_{x \to \infty} 5x \ln \left(1 + \frac{2}{x} \right)$

8. A company wants to construct a rectangular, six-sided cardboard box with as large of a volume as possible. There are two constraints on this box:

   - The length of its base must be three times the width. (The width of the box will be denoted by x and the height denoted by h. See the figure below.)
   - The box can only use 72ft^2 of cardboard. (In other words, the surface area of the box is limited to 72ft^2)



   (a) Write the height of the box h in terms of x.

   (b) Write a function $V(x)$ that expresses the volume of the box in terms of x.

   (c) Find the width of the box that gives the maximum possible volume. You must explicitly show that you have found a maximum. Do not simplify your answer.